THE MAXIMAL OPEN WHERE S HAS A TAU-LOCAL PROPERTY IS R-INVARIANT

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Allowed standard prerequisites (editorial).

- Tau-local properties of morphisms of schemes: notion of a property local on the target for the tau-topology (Descent chapter, Definition descent-definition-property-morphisms-local, canonical tag 02KO). cited at statement lines 655-656. Fixes the class of properties P for which the maximal open W is defined and the covering argument applies.
- R-invariant opens of a groupoid: W with t^{-1}(W) = s^{-1}(W) (Groupoids chapter, Definition groupoids-definition-invariant-open, canonical tag 03BC). cited at statement lines 660-661. States the target notion: W is R-invariant exactly when its inverse images under s and t agree.
- Largest open of the base on which a tau-locally checked morphism property holds, with its characterization over coverings (Descent chapter, Lemma descent-lemma-largest-open-of-the-base, canonical tag 06QP). cited at proof lines 665-674. Used twice: it gives the maximal open W for s|_{s^{-1}(W)} and, applied over the tau-coverings {s} and {t} with the fibre product square of the groupoid diagram, identifies t^{-1}(W) = W_1 = s^{-1}(W).

Source reference. Original reference 02KO: sources/stacks-chow/tags.txt, source lines 3379–3379. Cited at statement lines 655-656: P is a property of morphisms tau-local on the target.

Source reference. Original reference 03BC: sources/stacks-chow/tags.txt, source lines 4343–4343. Cited at statement lines 660-661: W is R-invariant, meaning t^{-1}(W) = s^{-1}(W).

Source reference. Original lemma 06QP: sources/stacks-chow/tags.txt, source lines 8613–8613. Cited at proof lines 665-666 and 671-674: existence and maximality of W, and the identification t^{-1}(W) = W_1 = s^{-1}(W) over the tau-coverings {s} and {t}.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid over S. Let $\tau \in \{Zariski, fppf, étale, smooth, syntomic\}$[1]. Let $\mathcal{P}$ be a property of morphisms*

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[1]The fact that $fpqc$ is missing is not a typo.

of schemes which is τ-local on the target (Descent, Definition 02KO). Assume $\{s : R \to U\}$ and $\{t : R \to U\}$ are coverings for the τ-topology. Let $W \subset U$ be the maximal open subscheme such that $s|_{s^{-1}(W)} : s^{-1}(W) \to W$ has property $\mathcal{P}$. Then W is R-invariant, see Groupoids, Definition 03BC.

(0.2.1)
$$\begin{array}{ccccc} & & U & & \\ & \overset{t}{\nearrow} & & \overset{t}{\nwarrow} & \\ R & \xleftarrow[\mathrm{pr}_0]{} & R \times_{s,U,t} R & \xrightarrow[c]{} & R \\ \downarrow{\scriptstyle s} & & \downarrow{\scriptstyle \mathrm{pr}_1} & & \downarrow{\scriptstyle s} \\ U & \xleftarrow{t} & R & \xrightarrow{s} & U \end{array}$$

Proof. The existence and properties of the open $W \subset U$ are described in Descent, Lemma 06QP. In Diagram (0.2.1) let $W_1 \subset R$ be the maximal open subscheme over which the morphism $\mathrm{pr}_1 : R \times_{s,U,t} R \to R$ has property $\mathcal{P}$. It follows from the aforementioned Descent, Lemma 06QP and the assumption that $\{s : R \to U\}$ and $\{t : R \to U\}$ are coverings for the τ-topology that $t^{-1}(W) = W_1 = s^{-1}(W)$ as desired. □